

Graphene coated nanospheres

We derive transmission coefficients, t_{TE}, t_{TM} reported in **Equations 6a and 6b** of the paper: [Localized plasmons in graphene-coated nanospheres](#)

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414>).

We first consider a bulk sphere without a graphene coating.

Bulk sphere without graphene coating

This set of notes I wrote will be helpful: <https://hackmd.io/@aligho/SymTGLcPxg>

(<https://hackmd.io/@aligho/SymTGLcPxg>).

We first consider TM modes, for which we have:

$\mathbf{H}(\mathbf{r}, \omega) = z_l(r\omega/c) \mathbf{L}Y_{lm}(\theta, \phi)$, where $z_l(z)$ is a spherical Bessel function of the first or second kinds or a spherical Hankel function. To make a connection to Thomas's paper, we note that $\nabla \times (\mathbf{r}\psi(\mathbf{r})) = \varepsilon_{ijk} \partial_j (r_k \psi(\mathbf{r})) = \varepsilon_{ijk} \delta_{jk} \psi(\mathbf{r}) + \varepsilon_{ijk} r_k \partial_j \psi(\mathbf{r}) = -\mathbf{r} \times \nabla \psi(\mathbf{r})$, which is just proportional to the angular momentum operator (note also that the angular momentum does not act on the radial coordinate r , which is why we were allowed to freely move $z_l(r\omega/c)$ to the left of the angular momentum operator).

We consider the incident field to be given by $\mathbf{H}^{\text{incident}}(\mathbf{r}, \omega) = j_l(r\omega/c) \mathbf{L}Y_{lm}(\theta, \phi)$ (a partial wave of an incoming plane wave), the internal field to be given by

$\mathbf{H}^{\text{internal}}(\mathbf{r}, \omega) = a j_l(nr\omega/c) \mathbf{L}Y_{lm}(\theta, \phi)$, and the scattered field to be given by

$\mathbf{H}^{\text{scattered}}(\mathbf{r}, \omega) = s h_l^1(r\omega/c) \mathbf{L}Y_{lm}(\theta, \phi)$

The matching conditions on the magnetic field require the following at the boundary of the sphere (i.e. at $r = R$):

$$j_l(r\omega/c) + s h_l^{(1)}(r\omega/c) = a j_l(nr\omega/c)$$

We now find the electric field, $\nabla \times \mathbf{H}(\mathbf{r}, \omega) = -i\omega\varepsilon_0 \mathbf{E}(\mathbf{r}, \omega)$

Therefore, we match the tangential component of the electric field by requiring the following at the sphere boundary:

$$a\partial_r(rj_l(rn\omega/c))|_{r=R} = \varepsilon \left[\partial_r(rj_l(r\omega/c)) + s\partial_r(rh_l^{(1)}(r\omega/c)) \right] |_{r=R}$$

By noting that, for a generic function, $f(z)$, and a generic constant, α , we have that $\partial_r(rf(\alpha r)) = f(\alpha r) + (\alpha r)f'(\alpha r) = \partial_x(xf(x))|_{x=\alpha r}$, we rewrite the boundary condition on the electric field as the following (defining $x \equiv ny, y \equiv r\omega/c$):

$$a\partial_x(xj_l(x)) = \varepsilon \left[\partial_y(yj_l(y)) + s\partial_y(yh_l^{(1)}(y)) \right]$$

We are ultimately trying to find the amplitude of the scattered field, which is given by s . To find this, we combine our two boundary conditions, giving us the following:

$$[j_l(y) + sh_l^{(1)}(y)]\partial_x(xj_l(x)) = \varepsilon \left[\partial_y(yj_l(y)) + s\partial_y(yh_l^{(1)}(y)) \right] j_l(x)$$

Which gives us:

$$s \left[\varepsilon j_l(x)(yh_l^{(1)}(y))' - h_l^{(1)}(y)(xj_l(x))' \right] = [j_l(y)(xj_l(x))' - \varepsilon(yj_l(y))'j_l(x)]$$

Therefore, we have:

$$t_{TM} \equiv s = \frac{j_l(y)(xj_l(x))' - \varepsilon(yj_l(y))'j_l(x)}{\varepsilon j_l(x)(yh_l^{(1)}(y))' - h_l^{(1)}(y)(xj_l(x))'}$$

which corresponds to **Equation 6b** of Thomas's paper in the limit $\sigma(\omega) \rightarrow 0$ (equivalently $g(\omega) \rightarrow 0$).

Magnetic (TE) modes

The matching conditions require $j_l(r\omega/c) + sh_l^{(1)}(r\omega/c) = aj_l(nr\omega/c)$ (tangential component of electric field) and

$$a\partial_r(rj_l(rn\omega/c)) = \left[\partial_r(rj_l(r\omega/c)) + s\partial_r(rh_l^{(1)}(r\omega/c)) \right]$$

Therefore, we get immediately:

$$t_{TE} \equiv s = \frac{[j_l(y)(xj_l(x))' - (yj_l(y))'j_l(x)]}{j_l(x)(yh_l^{(1)}(y))' - h_l^{(1)}(y)(xj_l(x))'},$$

which corresponds to **Equation 6a** in Thomas's paper in the limit $\sigma(\omega) \rightarrow 0$. Note that we may write this in a slightly different (and non-standard) way by expanding the derivatives and subsequently dividing the numerator and denominator by y :

$$t_{TE} = \frac{nj_l(y)j_l'(x) - j_l'(y)j_l(x)}{j_l(x)(h_l^{(1)}(y))' - nh_l^{(1)}(y)j_l'(x)}$$

With graphene (or any monolayer with finite surface impedance)

We now have a surface current, so $\mathbf{e}_r \times (\mathbf{H}_2 - \mathbf{H}_1) = \sigma(\omega)\mathbf{E}$

We first note, for future reference, that $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{e}_\phi$, $\mathbf{e}_r \times \mathbf{e}_\phi = -\mathbf{e}_\theta$

We again start with the TM modes:

We consider the incident field to be given by $\mathbf{H}^{\text{incident}}(\mathbf{r}, \omega) = j_l(r\omega/c)\mathbf{L}Y_{lm}(\theta, \phi)$, the internal field to be given by $\mathbf{H}^{\text{internal}}(\mathbf{r}, \omega) = aj_l(rn\omega/c)\mathbf{L}Y_{lm}(\theta, \phi)$, and the scattered field to be given by $\mathbf{H}^{\text{scattered}}(\mathbf{r}, \omega) = sh_l^1(r\omega/c)\mathbf{L}Y_{lm}(\theta, \phi)$.

$$\mathbf{E}(\mathbf{r}, \omega) = (i/\omega\epsilon_0\epsilon)\nabla \times \mathbf{H}(\mathbf{r}, \omega) \rightarrow E_\phi(\mathbf{r}, \omega) = (i/\omega\epsilon_0\epsilon)\frac{1}{r}\partial_r(rH_\theta(\mathbf{r}, \omega)), \text{ and}$$

$$E_\theta(\mathbf{r}, \omega) = -(i/\omega\epsilon_0\epsilon)\frac{1}{r}\partial_r(rH_\phi(\mathbf{r}, \omega)) \text{ because } H_r = 0 \text{ for TM modes}$$

Therefore, we have:

$$j_l(r\omega/c) + sh_l^1(r\omega/c) - aj_l(nr\omega/c) = a\frac{i\sigma}{\omega\epsilon_0\epsilon R}\partial_x(xj_l(x)) = a(g(\omega)/\epsilon)\partial_x(xj_l(x)),$$

where $g(\omega)$ is defined in **Equation 7** of the Christensen paper. Additionally, we have (continuity of the tangential component of the electric field):

$$a\partial_x(xj_l(x)) = \epsilon\left[\partial_y(yj_l(y)) + s\partial_y(yh_l(y))\right]$$

Therefore, putting both boundary conditions gives us:

$$\partial_x(xj_l(x)) \frac{[j_l(y) + sh_l^{(1)}(y)]}{j_l(x) + (g(\omega)/\varepsilon)\partial_x(xj_l(x))} = \varepsilon [\partial_y(yj_l(y)) + s\partial_y(yh_l^{(1)}(y))]$$

By putting all terms proportional to the scattered field amplitude on one side of the equation, we readily obtain:

$$s \left[(xj_l(x))' h_l^{(1)}(y) - \varepsilon j_l(x) (yh_l^{(1)}(y))' - g(\omega) (xj_l(x))' (yh_l^{(1)}(y))' \right] = \varepsilon j_l(x) (yj_l(y))' + g(\omega) (yj_l(y))' (xj_l(x))' - j_l(y) (xj_l(x))'$$

We write this more explicitly as:

$$s = \frac{\varepsilon j_l(x) (yj_l(y))' + g(\omega) (yj_l(y))' (xj_l(x))' - j_l(y) (xj_l(x))'}{(xj_l(x))' h_l^{(1)}(y) - \varepsilon j_l(x) (yh_l^{(1)}(y))' - g(\omega) (xj_l(x))' (yh_l^{(1)}(y))'},$$

which is **equation 6b** in the Christensen paper.

TE modes with graphene

The matching conditions require $j_l(r\omega/c) + sh_l^{(1)}(r\omega/c) = aj_l(nr\omega/c)$ (tangential component of electric field). And we must also have:

$$e_r \times (\nabla \times \mathbf{E}_2 - \nabla \times \mathbf{E}_1) = i\omega\sigma(\omega)\mu_0 \mathbf{E}$$

Which gives us:

$$-(yj_l(y))' - s(yh_l^{(1)}(y))' + a(xj_l(x))' = ai\omega\sigma\mu_0 Rj_l(x) = ag(\omega)(R\omega/c)^2 j_l(x)$$

From this we get:

$$j_l(x) (yj_l(y))' + sj_l(x) (yh_l^{(1)}(y))' = \left[-g(\omega)(R\omega/c)^2 j_l(x) + (xj_l(x))' \right] [j_l(y) + sh_l^{(1)}(y)]$$

From this we obtain:

$$s \left[j_l(x) (yh_l^{(1)}(y))' - h_l^{(1)}(y) (xj_l(x))' + (R\omega/c)^2 g(\omega) j_l(x) h_l^{(1)}(y) \right] = j_l(y) (xj_l(x))' - j_l(x) (yj_l(y))' - (R\omega/c)^2 g(\omega) j_l(x) j_l(y)$$

We may write this more explicitly as:

$$s = \frac{j_l(y)(xj_l(x))' - j_l(x)(yj_l(y))' - (R\omega/c)^2 g(\omega)j_l(x)j_l(y)}{j_l(x)(yh_l^{(1)}(y))' - h_l^{(1)}(y)(xj_l(x))' + (R\omega/c)^2 g(\omega)j_l(x)h_l^{(1)}(y)},$$

which is **Equation 6a** in the Christensen paper.

Plasmon modes

The plasmon modes, in the quasistatic approximation, given by **Equation 10** in the Christensen paper are given, for angular momentum, l ,

$$l\varepsilon_1 + (l+1)\varepsilon_2 + l(l+1)g(\omega) = 0$$

where $g(\omega) = i\sigma(\omega)/(\omega\varepsilon_0 R)$. Note that $\sigma(\omega)$ is a surface conductivity, meaning that it gives a surface current density, which is why $g(\omega)$ is a dimensionless quantity.

We rewrite this as (in the nomenclature of the Christensen paper):

$$g(\omega) = \frac{ie^2}{\pi(\hbar\omega)\varepsilon_0 R} \left[\sigma_{\text{intra}}/\sigma_0 + \sigma_{\text{inter}}/\sigma_0 \right] = \frac{4i(\hbar c\alpha)}{(\hbar\omega)R} \left[\sigma_{\text{intra}}/\sigma_0 + \sigma_{\text{inter}}/\sigma_0 \right],$$

where we used the fact that $e^2/4\pi\varepsilon_0\hbar c = \alpha$, where $\alpha = 1/137$ is the fine structure constant and $\hbar c = 6.6 \times 3 \times 10 \approx 200(\text{nm} \times \text{eV})$ and used the quantity $\sigma_0 = e^2/\pi\hbar$, which is introduced in the Christensen paper.

In the low temperature limit, neglecting the interband contribution, we have:

$$\sigma_{\text{intra}}/\sigma_0 = \frac{2ik_B T}{\hbar\omega} \ln(2 \cosh(\varepsilon_F/2k_B T)) \approx \frac{2ik_B T}{\hbar\omega} \frac{\varepsilon_F}{2k_B T} = \frac{i\varepsilon_F}{\hbar\omega},$$

where we used the fact that, for $\varepsilon_F \gg k_B T$, we have

$2 \cosh(\varepsilon_F/2k_B T) \approx \exp(\varepsilon_F/2k_B T)$. Note that this means that in the low temperature, intraband, approximation, we have $\sigma(\omega) = ie^2\varepsilon_F/\pi\hbar(\hbar\omega)$, which is consistent with, e.g.,

Plasmonics in graphene at infrared frequencies

(<https://journals.aps.org/prb/abstract/10.1103/PhysRevB.80.245435>). Therefore, in the **low temperature, intraband regime**, we have:

$$g(\omega) \approx -\frac{800\alpha\varepsilon_F}{(\hbar\omega)^2 R}$$

where R is given in nanometers, ε_F , $\hbar\omega$ are given in eV (this must be so because the numerical prefactor of ≈ 800 is given in $\text{nm} \times \text{eV}$). Therefore, the dispersion is given by:

$$l\varepsilon_1 + (l+1)\varepsilon_2 - l(l+1)\frac{800\alpha\varepsilon_F}{(\hbar\omega)^2 R} = 0 \rightarrow \hbar\omega/\varepsilon_F = \frac{1}{\varepsilon_F} \left[\frac{800l(l+1)\alpha\varepsilon_F}{R[l\varepsilon_1 + (l+1)\varepsilon_2]} \right]^{1/2}$$

Let's see what this looks like for the case, $\varepsilon_1 = \varepsilon_2 = 1$ (which corresponds to a vacuum bulk and vacuum environment separated by a coating):

$$\hbar\omega/\varepsilon_F = \frac{1}{\varepsilon_F} \left[\frac{800\alpha\varepsilon_F}{R} \right]^{1/2} \left[\frac{l(l+1)}{2l+1} \right]^{1/2}$$

Therefore, for a fixed value of the Fermi energy, ε_F , and a fixed of the scatterer radius, R , the plasmon mode scales as $\sqrt{(l^2 + l)/(2l + 1)}$. Meaning that the ratio of plasmon frequencies corresponding to the l 'th and m 'th angular momenta is given by:

$$\omega_l/\omega_m = \sqrt{[(l^2 + l)(2m + 1)]/[(m^2 + m)(2l + 1)]}$$

As a sanity check, we calculate this for a particular case:

$\omega_5/\omega_1 = \sqrt{(30 \times 3)/(2 \times 11)} \approx 2$, which is consistent with **Figure 2(a)** in the Christensen paper.

Interband contribution

We now include the interband contribution into $g(\omega)$. We first specialize to zero temperature, where the result may be found analytically. We don't use **Equation 12(b)** of the paper. Instead, we refer to the paper: Dynamical polarization of graphene at finite doping (<https://iopscience.iop.org/article/10.1088/1367-2630/8/12/318/meta>). Weirdly, the relevant equation in this paper is also **Equation 12**. That equation gives us the polarization function, $P(\mathbf{q}, \omega)$ which gives us the induced electron density as a response to the scalar potential. The scalar potential is related to the electric field via, $\mathbf{E}(\mathbf{q}, \omega) = -i\mathbf{q}\phi(\mathbf{q}, \omega)$ and the

induced charge density is related to the current via $\omega\rho(\mathbf{q}, \omega) = \mathbf{q} \cdot \mathbf{j}(\mathbf{q}, \omega)$. Therefore, we have that $\left[|\mathbf{q}|/\omega\right]j(\mathbf{q}, \omega) = \left[ie^2/|\mathbf{q}|\right]P(\mathbf{q}, \omega)\mathbf{E}(\mathbf{q}, \omega)$. From which we read off the relation between the longitudinal conductivity and the polarization function:

$$\sigma(\mathbf{q}, \omega) = ie^2\omega P(\mathbf{q}, \omega)/|\mathbf{q}|^2.$$

The interband contribution is of the form:

$$\sigma_{\text{inter}}/\sigma_0 = \left[\pi\hbar/e^2\right] \left[ie^2\omega/|\mathbf{q}|^2\right] \left[|\mathbf{q}|^2/4\pi\hbar\omega\right] \left[\ln \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega}\right] = \frac{i}{4} \ln \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega}$$

Combining this with the intraband result gives us:

$$g(\omega) = -\frac{4(\hbar c\alpha)}{(\hbar\omega)R} \left[\frac{\varepsilon_F}{\hbar\omega} + \frac{1}{4} \ln \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega} \right],$$

$$\sigma(\omega) = i4\alpha(c\varepsilon_0) \left[\frac{\varepsilon_F}{\hbar\omega} + \frac{1}{4} \ln \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega} \right]$$

So, the dispersion is given by:

$$\frac{\varepsilon_1}{l+1} + \frac{\varepsilon_2}{l} = \frac{4(\hbar c\alpha)}{(\hbar\omega)R} \left[\frac{\varepsilon_F}{\hbar\omega} + \frac{1}{4} \ln \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega} \right]$$

For $\varepsilon_{1,2} > 0$ (dielectric sphere and environment), the right hand side of this equation has to be positive for a solution to exist. Let's find the region where this is satisfied. Taking $x \equiv \varepsilon_F/\hbar\omega$, we find that we must require the following condition to hold:

$$4x > \ln (2x+1)/(2x-1) \rightarrow \frac{2x+1}{2x-1} < e^{4x}. \text{ At } x=1, \text{ this is clearly satisfied. At}$$

$x=1/4$, the RHS is e and the LHS is equal to 3, so this is almost the crossover point.

Hydrodynamic corrections in the coating

The hydrodynamic corrections change the current-field relationship to:

$$(\beta^2/\omega^2)\nabla_{||}(\nabla_{||} \cdot \mathbf{j}(\mathbf{r}, \omega)) + \mathbf{j}(\mathbf{r}, \omega) = \sigma(\omega)\mathbf{E}(\mathbf{r}, \omega),$$

where we have introduced the tangential gradient, which acts only on the boundary, $r = R$, and measures only changes in the polar and azimuthal directions:

$$\nabla_{||} \equiv R^{-1} \left[e_{\theta} \partial_{\theta} + \frac{1}{\sin(\theta)} e_{\phi} \partial_{\phi} \right]$$

We write this in terms of the charge density, $\nabla \cdot \mathbf{j} = i\omega\rho$ by taking the divergence of both sides of the equation and using $\mathbf{E}(\mathbf{r}, \omega) = -\nabla\phi(\mathbf{r}, \omega)$, in the quasistatic approximation. Explicitly, we mean the following:

$$\begin{aligned} (\beta^2/\omega^2)\nabla_{||}(\nabla_{||} \cdot \mathbf{j}(\mathbf{r}, \omega)) + \mathbf{j}(\mathbf{r}, \omega) = \sigma\mathbf{E}(\mathbf{r}, \omega) &\rightarrow (\beta^2/\omega^2)\nabla_{||}^2(\nabla_{||} \cdot \mathbf{j}(\mathbf{r}, \omega)) + \nabla_{||} \cdot \mathbf{j}(\mathbf{r}, \omega) \\ &= \sigma(\omega)\nabla_{||} \cdot \mathbf{E}(\mathbf{r}, \omega) \end{aligned}$$

Using the continuity relation mentioned above and the relation between the electric field and scalar potentials then gives us:

$$(\beta^2/\omega^2)\nabla_{||}^2(i\omega\rho(\mathbf{r}, \omega)) + i\omega\rho(\mathbf{r}, \omega) = -\sigma(\omega)\nabla_{||}^2\phi(\mathbf{r}, \omega),$$

We divide both sides by $i\omega$ to give us:

$$\rho(\mathbf{r}, \omega) + \frac{\beta^2}{\omega^2}\nabla_{||}^2\rho(\mathbf{r}, \omega) = \frac{i\sigma(\omega)}{\omega}\nabla_{||}^2\phi_{||}(\mathbf{r}, \omega),$$

which is **Equation A1** of the **Appendix**. We expand the potential, $\phi(\mathbf{r}, \omega)$ in terms of different angular momentum numbers (noting that the potential must satisfy Laplace's equation everywhere but the boundary):

$$\phi(\mathbf{r}) = (r^l + br^{-l-1})P_l(\cos(\theta)) \text{ for } r > R \text{ and } \phi(\mathbf{r}) = ar^lP_l(\cos(\theta)) \text{ for } r < R$$

Our boundary conditions require the following two conditions to hold:

$$\begin{aligned} R^l + bR^{-l-1} &= aR^l \rightarrow b = R^{2l+1}(a - 1) \text{ as well as} \\ -\varepsilon_2 \left[lR^{l-1} - b(l+1)R^{-l-2} \right] + \varepsilon_1 \left[alR^{l-1} \right] &= \rho_l/\varepsilon_0 \end{aligned}$$

Now, note that the tangential laplacian will be given by (in the case where we are dealing with azimuthally symmetric scalar functions, i.e. $\partial_\phi \rightarrow 0$):

$$\nabla_{||}^2 = \frac{1}{R^2 \sin(\theta)} \partial_\theta (\sin(\theta) \partial_\theta f)$$

In our case, we are concerned with functions of the form $f = f(\cos(\theta))$, for which we have $\sin(\theta) \partial_\theta f = -\sin^2(\theta) f' = -(1 - \cos(\theta))^2 f'$

Therefore, the tangential Laplacian may be written as:

$$\nabla_{||}^2 \rightarrow \frac{1}{R^2} \partial_x ((1 - x^2) f'(x))$$

For $f = P_l(\cos(\theta))$ this gives us $-l(l+1)$, from the well known Legendre differential equation.

Therefore, we may write:

$$\rho_l - \frac{\beta^2}{R^2 \omega^2} l(l+1) \rho_l = -\frac{i\sigma(\omega)}{R^2 \omega} l(l+1) a R^l \rightarrow$$

$$\rho_l (1 - \beta^2 l(l+1)/R^2 \omega^2) = -i\sigma(\omega) l(l+1) a / R^{2-l} \omega$$

Simplified,

$$\rho_l = -ia \frac{\sigma(\omega) l(l+1)}{R^{2-l} \omega} \frac{1}{1 - \beta^2 l(l+1)/R^2 \omega^2}$$

Which is **Equation A3a (of the Appendix)**.

We now seek to derive **Equation A4 (in the Appendix)**. To do this, we use the additional equation, from combining the boundary conditions above:

$$-\varepsilon_2 \left[l - (a-1)(l+1) \right] + \varepsilon_1 \left[al \right] = \rho_l R^{1-l} / \varepsilon_0 = -ia \frac{\sigma(\omega) l(l+1)}{\varepsilon_0 R \omega} \frac{1}{1 - \beta^2 l(l+1)/R^2 \omega^2}$$

We write this in terms of the hydrodynamic corrected $g(\omega)$:

$$\begin{aligned}
-\varepsilon_2 \left[l - (a-1)(l+1) \right] + \varepsilon_1 \left[al \right] &= -ag(\omega)l(l+1) \rightarrow \\
a \left[l\varepsilon_1 + (l+1)\varepsilon_2 \right] - (2l+1)\varepsilon_2 &= -ag(\omega)l(l+1) \rightarrow \\
a \left[l\varepsilon_1 + (l+1)\varepsilon_2 + l(l+1)g(\omega) \right] &= \varepsilon_2(2l+1)
\end{aligned}$$

Therefore, $b = R^{2l+1}(a-1)$, which corresponds to the amplitude of the scattered transverse magnetic polarized wave, is given by:

$$R^{2l+1} \left[\frac{l\varepsilon_2 - l\varepsilon_1 - l(l+1)g(\omega)}{(l+1)\varepsilon_2 + l\varepsilon_1 + l(l+1)g(\omega)} \right], \text{ which exactly gives **Equation A4** of the}$$

Appendix.

Appendix 1: Equation 12 of Christensen paper

In the above, we used the analytic formula for the zero temperature result. Here, we show how this result may be found from the general result **Equation 12(b)** of Thomas's paper. We focus first on the real part of the conductivity, $\sigma_{\text{inter}}/\sigma_0$, since this just becomes a step function. This reads,

$$\sigma_{\text{inter}}/\sigma_0 = \frac{\pi}{4} H(\hbar\omega/2) + \Re \left[i\hbar(\omega + i\delta) \int_0^\infty \frac{H(\varepsilon) - H(\hbar\omega/2)}{\hbar^2(\omega + i\delta)^2 - 4\varepsilon^2} d\varepsilon \right],$$

where

$$H(\varepsilon) = \frac{\sinh(\varepsilon/k_B T)}{\cosh(\varepsilon/k_B T) + \cosh(\varepsilon_F/k_B T)}$$

We focus first on the first term in the low temperature limit, $\hbar\omega \gg k_B T, \varepsilon_F \gg k_B T$, which gives us:

$$\frac{\pi}{4} \frac{e^{\hbar\omega/2k_B T}}{e^{\hbar\omega/2k_B T} + e^{\varepsilon_F/k_B T}} = \frac{\pi}{4} \frac{1}{1 + e^{(\varepsilon_F - \hbar\omega/2)/k_B T}} = \frac{\pi}{4} \theta(\hbar\omega/2 - \varepsilon_F)$$

Note that this is basically just a Fermi function.

We now evaluate the second term:

$$\Re \left[i\hbar(\omega + i\delta) \int_0^\infty \frac{H(\varepsilon) - H(\hbar\omega/2)}{\hbar^2(\omega + i\delta)^2 - 4\varepsilon^2} d\varepsilon, \right]$$

To evaluate this, we decompose the expression into two denominators, as follows:

$$\frac{i\hbar(\omega + i\delta)}{\hbar^2(\omega + i\delta)^2 - 4\varepsilon^2} = \frac{i}{2} \left[\frac{1}{\hbar(\omega + i\delta) - 2\varepsilon} + \frac{1}{\hbar(\omega + i\delta) + 2\varepsilon} \right]$$

Note that we are only concerned with the real part of this expression (for now), which reads:

$$\frac{\pi}{2} \left[\delta(\hbar\omega - 2\varepsilon) + \delta(\hbar\omega + 2\varepsilon) \right] = \frac{\pi}{4} \left[\delta(\varepsilon - \hbar\omega/2) + \delta(\varepsilon + \hbar\omega/2) \right]$$

Therefore, we have:

$$\begin{aligned} & \Re \left[i\hbar(\omega + i\delta) \int_0^\infty \frac{H(\varepsilon) - H(\hbar\omega/2)}{\hbar^2(\omega + i\delta)^2 - 4\varepsilon^2} d\varepsilon = \right. \\ & \left. \int_0^\infty \frac{\pi}{4} \left[H(\varepsilon) - H(\hbar\omega/2) \right] \left[\delta(\varepsilon - \hbar\omega/2) + \delta(\varepsilon + \hbar\omega/2) \right] d\varepsilon = 0 \right. \end{aligned}$$

In the last equality, we used the fact that the integral only runs over positive energies, meaning that the second delta function does not contribute.

Therefore, we have that:

$$\sigma_{\text{inter}}/\sigma_0 = \frac{\pi}{4} \theta(\hbar\omega/2 - \varepsilon_F) \rightarrow \sigma_{\text{inter}} = \frac{e^2}{4\hbar} \theta(\hbar\omega/2 - \varepsilon_F)$$

From this, we may read off the contribution to the imaginary part of the polarization,

$$\left(-i|\mathbf{q}|^2/e^2\omega \right) \frac{e^2}{4\hbar} \theta(\hbar\omega/2 - \varepsilon_F) = \left[-i|\mathbf{q}|^2/(4\hbar\omega) \right] \theta(\hbar\omega/2 - \varepsilon_F)$$

Appendix 2: Finite temperature effects in Equation 12

Note that finite temperature effects do not change the fact that the second term in $\sigma_{\text{inter}}/\sigma_0$ does not contribute any real valued component.

Therefore, we are left with:

$$\Re(\sigma_{\text{inter}}(\hbar\omega)/\sigma_0) = \frac{\pi}{4} \frac{1}{1 + e^{(\varepsilon_F - \hbar\omega/2)/k_B T}} = \frac{\pi}{4} H(\hbar\omega/2)$$

Note that since for Kramers-Kronig relations (which will be used below) we also need the conductivity at negative frequencies and since $\sigma^*(\omega) = \sigma(-\omega)$, we have that

$$\Re(\sigma_{\text{inter}}(\hbar\omega)/\sigma_0) = \frac{\pi}{4} H(|\hbar\omega/2|)$$

and

$$\Im(\sigma_{\text{inter}}/\sigma_0) = \Re \left[\frac{1}{2} \int_0^\infty [H(\varepsilon) - H(\hbar\omega/2)] \left[\frac{1}{\hbar(\omega + i\delta) - 2\varepsilon} + \frac{1}{\hbar(\omega + i\delta) + 2\varepsilon} \right] d\varepsilon \right]$$

We now note that $H(-\varepsilon) = -H(\varepsilon)$. Therefore, in the second term in the above expression (corresponding to the second term in the second bracket), we take $u = -\varepsilon$ in order to write it as follows:

$$\Re \frac{1}{2} \int_{-\infty}^0 \left[-H(\varepsilon) - H(\hbar\omega/2) \right] \left[\frac{1}{\hbar(\omega + i\delta) - 2\varepsilon} \right] d\varepsilon$$

Note that the term proportional to $H(\hbar\omega/2)$ will now vanish when the first and second terms in our expression for $\Im(\sigma_{\text{inter}}/\sigma_0)$ are combined. So we are left with:

$$\begin{aligned} \Im(\sigma_{\text{inter}}/\sigma_0) = \Re \left[\frac{1}{4} \int_0^\infty \frac{H(\varepsilon/2)}{\hbar(\omega + i\delta) - \varepsilon} d\varepsilon - \frac{1}{4} \int_{-\infty}^0 \frac{H(\varepsilon/2)}{\hbar(\omega + i\delta) - \varepsilon} d\varepsilon \right] = \\ \Re \frac{1}{\pi} \int_{-\infty}^\infty \frac{\Re\sigma(\varepsilon)/\sigma_0}{\hbar\omega + i\hbar\delta - \varepsilon} d\varepsilon \end{aligned}$$

This is simply a statement of the Kramers-Kronig relations since

$$\int_{-\infty}^{\infty} \frac{\sigma(\varepsilon)}{\varepsilon - \hbar\omega + i\delta} d\varepsilon = 0 \text{ (closing the contour in the upper half of the complex plane)}$$

Therefore, we have that:

$$\sigma(\hbar\omega) = -\frac{i}{\pi} \int_{-\infty}^{\infty} \sigma(\varepsilon) \mathcal{R} \frac{1}{\varepsilon - \hbar\omega + i\delta} d\varepsilon$$

Let's verify that in the zero temperature case this all works out.

$$\begin{aligned} \Im(\sigma_{\text{inter}}/\sigma_0) &= -\frac{1}{4} \left[\int_{2\varepsilon_F}^{\infty} \left[\frac{1}{\varepsilon - \hbar\omega} \right] d\varepsilon + \int_{-\infty}^{-2\varepsilon_F} \left[\frac{1}{\varepsilon - \hbar\omega} \right] d\varepsilon \right] = \\ &= -\frac{1}{4} \log \frac{\infty}{2\varepsilon_F - \hbar\omega} \frac{2\varepsilon_F + \hbar\omega}{\infty} \end{aligned}$$

Which gives us:

$$\Im\sigma(\hbar\omega) = \frac{1}{4} \log \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega},$$

which is exactly what we had derived before from using the polarizability.